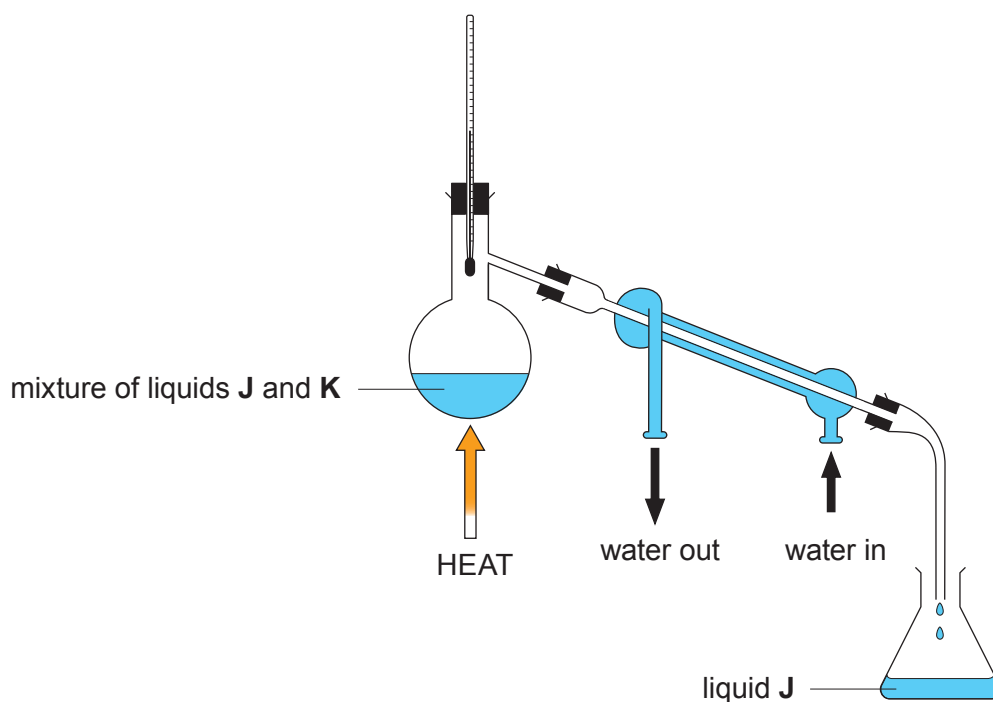


3. The diagram shows the separation of a mixture of liquids **J** and **K**.



- The temperature on the thermometer stays at 56°C even though the mixture is still being heated
- Drops of liquid **J** fall steadily into the flask

(a) Use the information given above and in the diagram.

State the conclusions that you can draw about the boiling points of liquids **J** and **K**. [2]

.....

.....

.....



- (b) **K** is a compound with the formula $\text{C}_4\text{H}_8\text{O}$. The relative formula mass (M_r) of compound **K** is 72.

Calculate the percentage by mass of carbon in compound **K**.

[2]

$$A_r(\text{C}) = 12$$

Percentage = %

4

